



**RAJALAKSHMI
ENGINEERING COLLEGE**

An AUTONOMOUS Institution
Affiliated to ANNA UNIVERSITY, Chennai

**DEPARTMENT OF COMPUTER SCIENCE AND
ENGINEERING LAB MANUAL**

CS23432 – Software Construction

(REGULATION 2023)

RAJALAKSHMI ENGINEERING COLLEGE

Thandalam, Chennai-602015

Name:

Register No:

Year / Branch / Section:

Semester:

Academic Year:

INDEX

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1.		Azure Devops Environment Setup.
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EXP NO: 1

AZURE DEVOPS ENVIRONMENT SETUP

Aim:

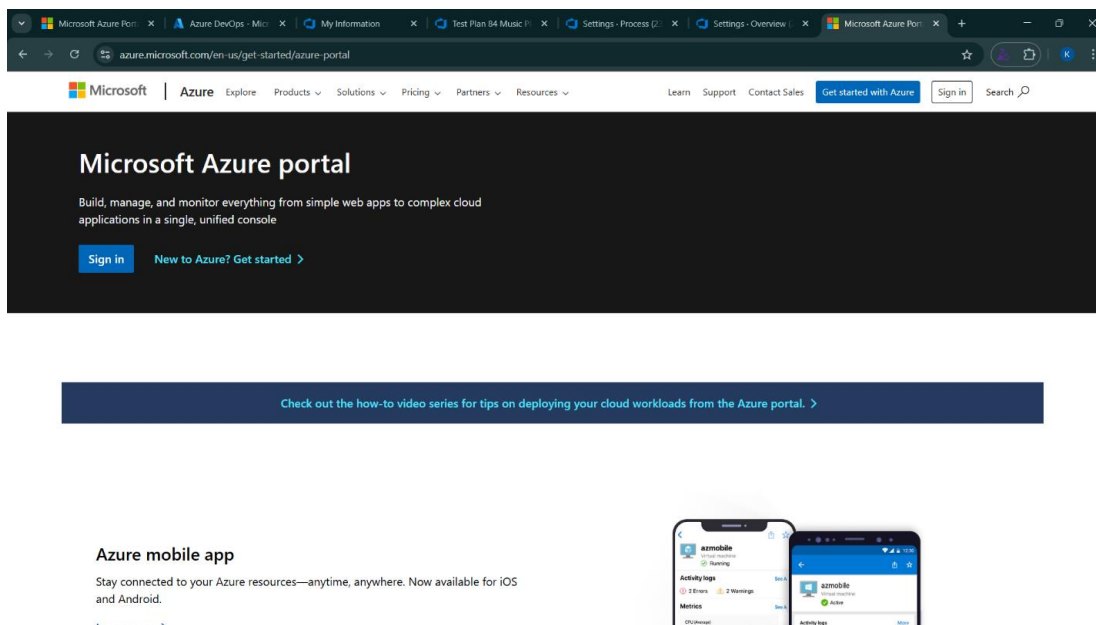
To set up and access the Azure DevOps environment by creating an organization through the Azure portal.

INSTALLATION

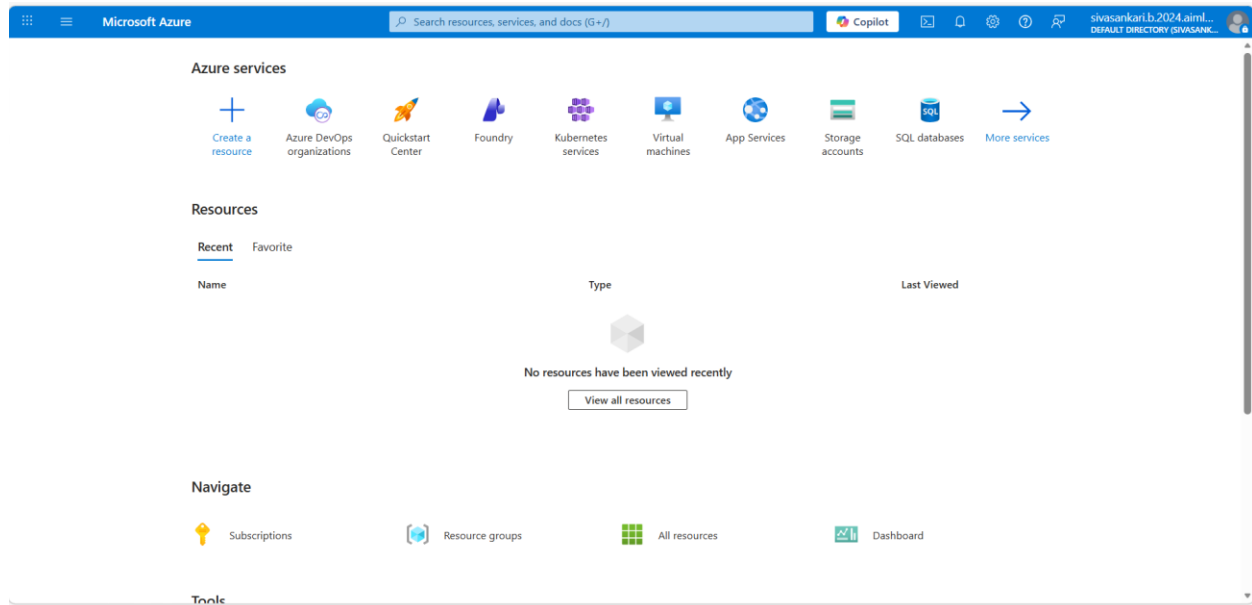
1. Open your web browser and go to the Azure website: <https://azure.microsoft.com/en-us/get-started/azure-portal>.

Sign in using your Microsoft account credentials.

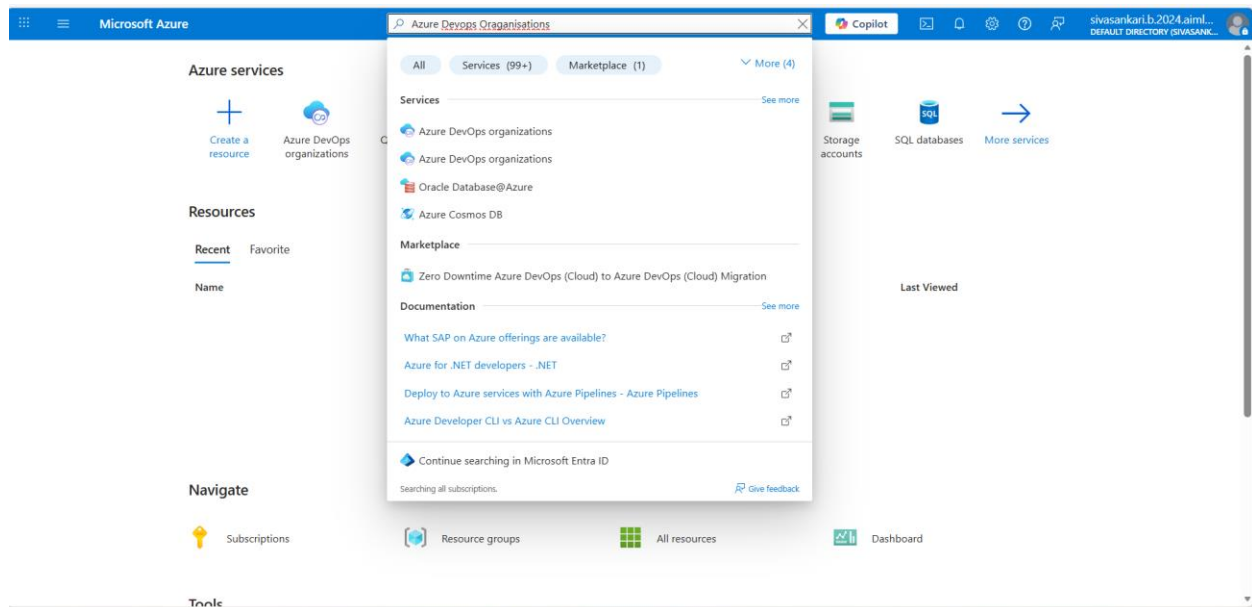
If you don't have a Microsoft account, you can create one here: <https://signup.live.com/?lic=1>



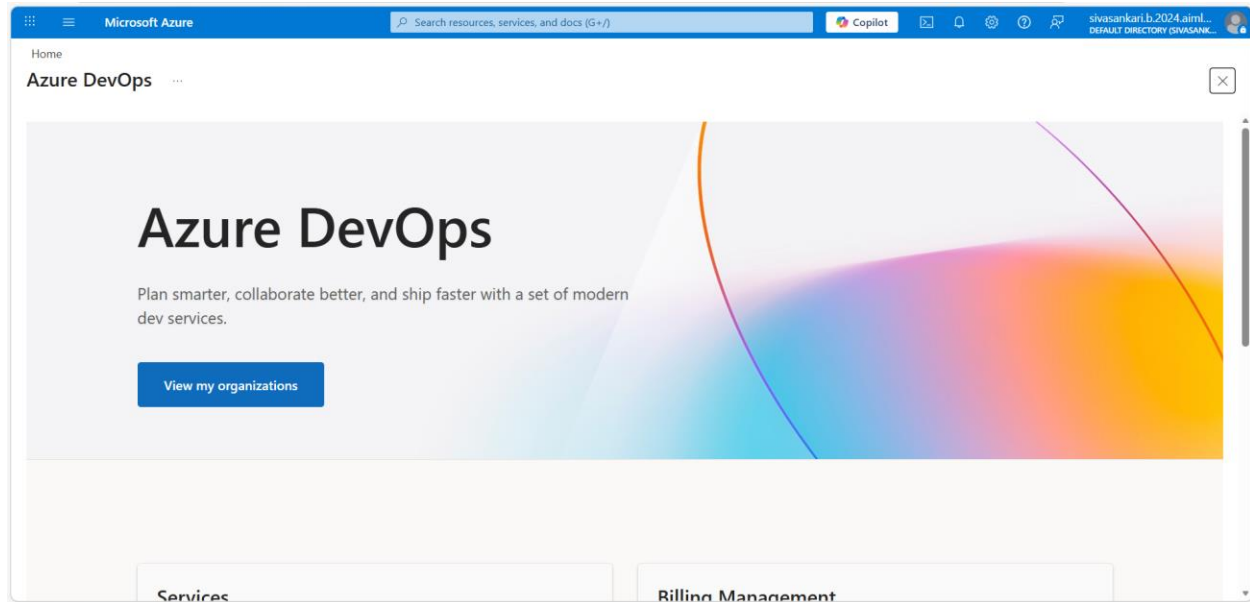
2. Azure home page



3. Open DevOps environment in the Azure platform by typing *Azure DevOps Organizations* in the search bar.



4. Click on the *My Azure DevOps Organization* link and create an organization and you should be taken to the Azure DevOps Organization Home page.



Result:

Successfully accessed the Azure DevOps environment and created a new organization through the Azure portal.

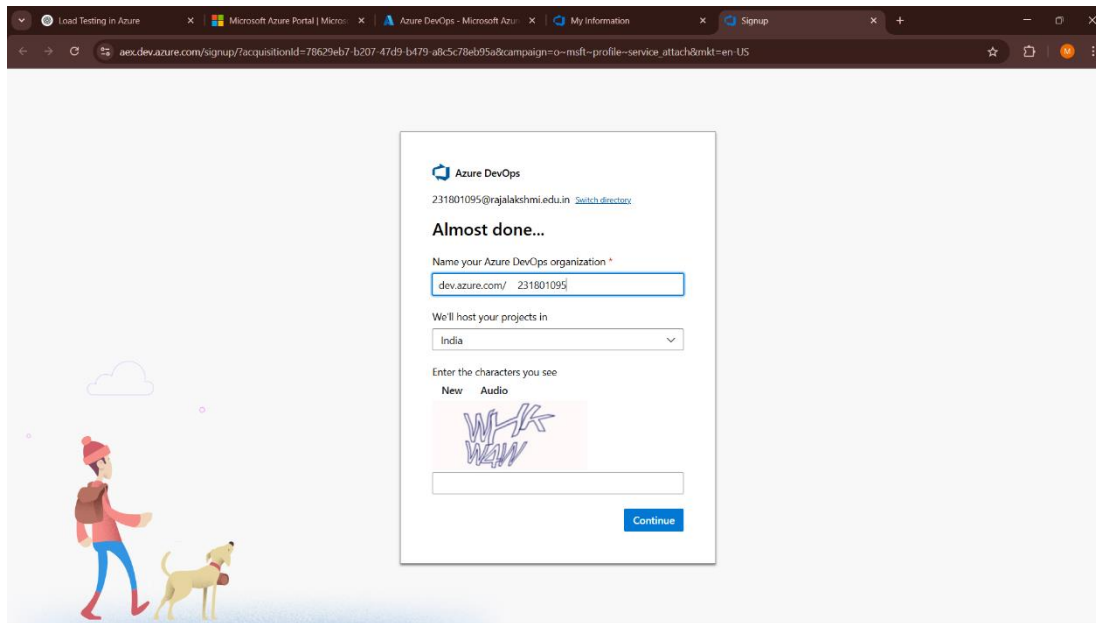
EXP NO: 2

AZURE DEVOPS PROJECT SETUP AND USER STORY MANAGEMENT

Aim:

To set up an Azure DevOps project for efficient collaboration and agile work management.

1.Create An Azure Account



2.Create the First Project in Your Organization

- After the organization is set up, you'll need to create your first **project**. This is where you'll begin to manage code, pipelines, work items, and more.
- On the organization's **Home page**, click on the **New Project** button.
- Enter the project name, description, and visibility options:
 - Name:** Choose a name for the project (e.g., **LMS**).
 - Description:** Optionally, add a description to provide more context about the project.
 - Visibility:** Choose whether you want the project to be **Private** (accessible only to those invited) or **Public** (accessible to anyone).
- Once you've filled out the details, click **Create** to set up your first project.

Create new project



Project name *

Smart-Building-Energy-Forecasting

Description

Predict Building energy consumption using ML

Visibility

 **Private**

Only people you give access to will be able to view this. Want to create a public project? [Try GitHub](#)

^ **Advanced**

Version control ?

Git

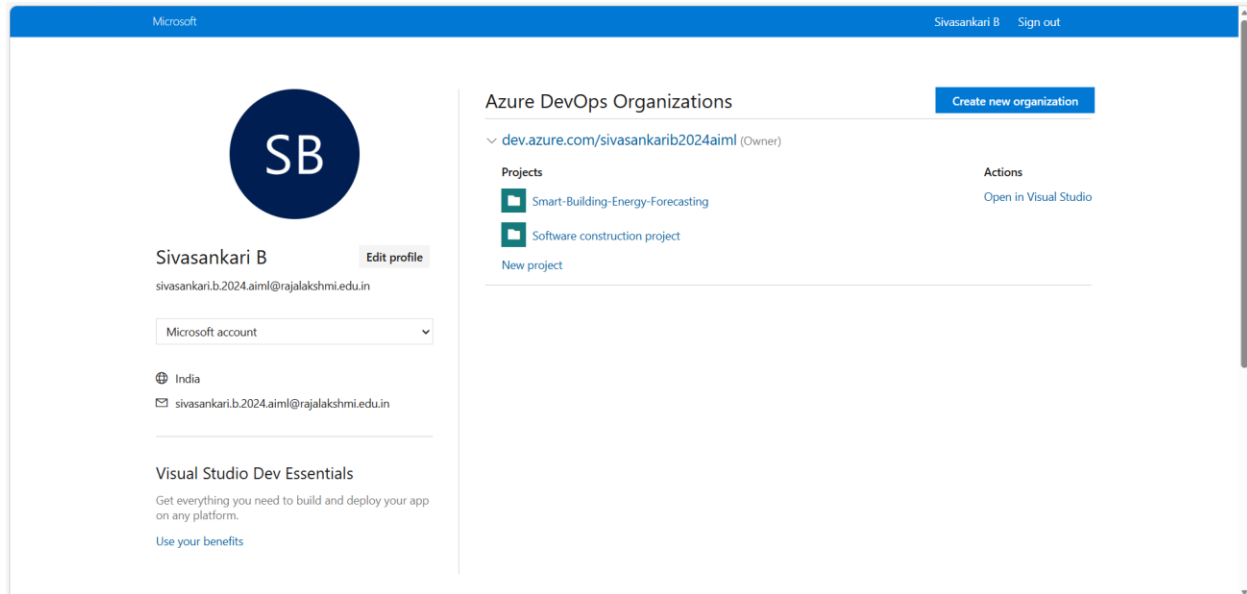
Work item process ?

Agile

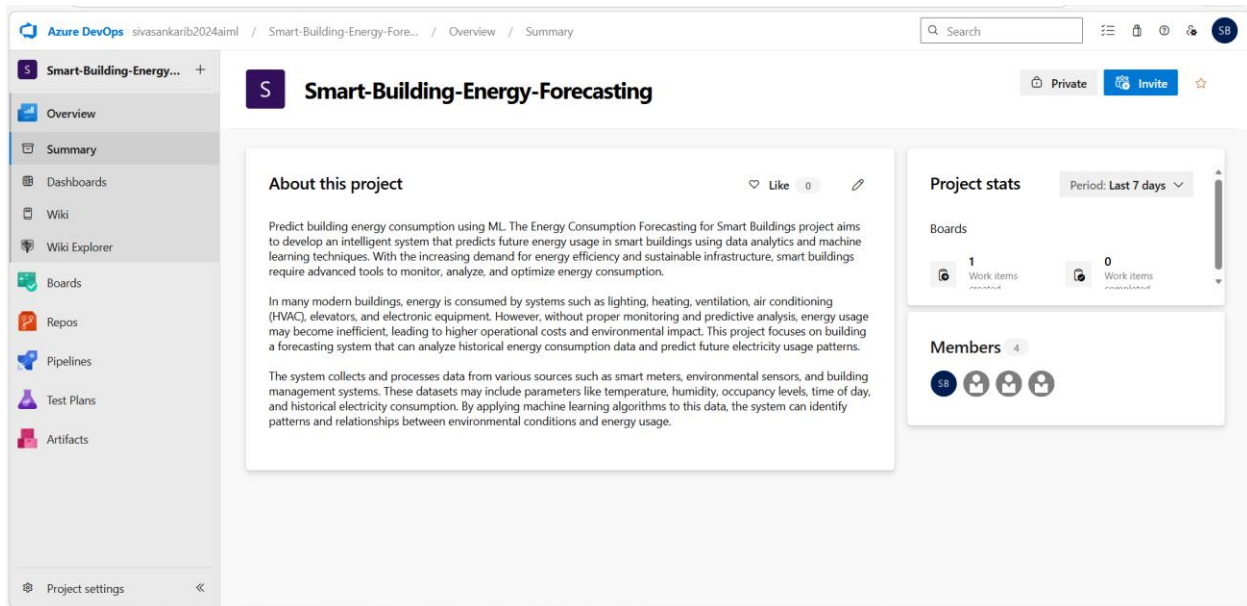
Cancel

Create

3. Once logged in, ensure you are in the correct organization. If you're part of multiple organizations, you can switch between them from the top left corner (next to your user profile). Click on the Organization name, and you should be taken to the Azure DevOps Organization Home page.

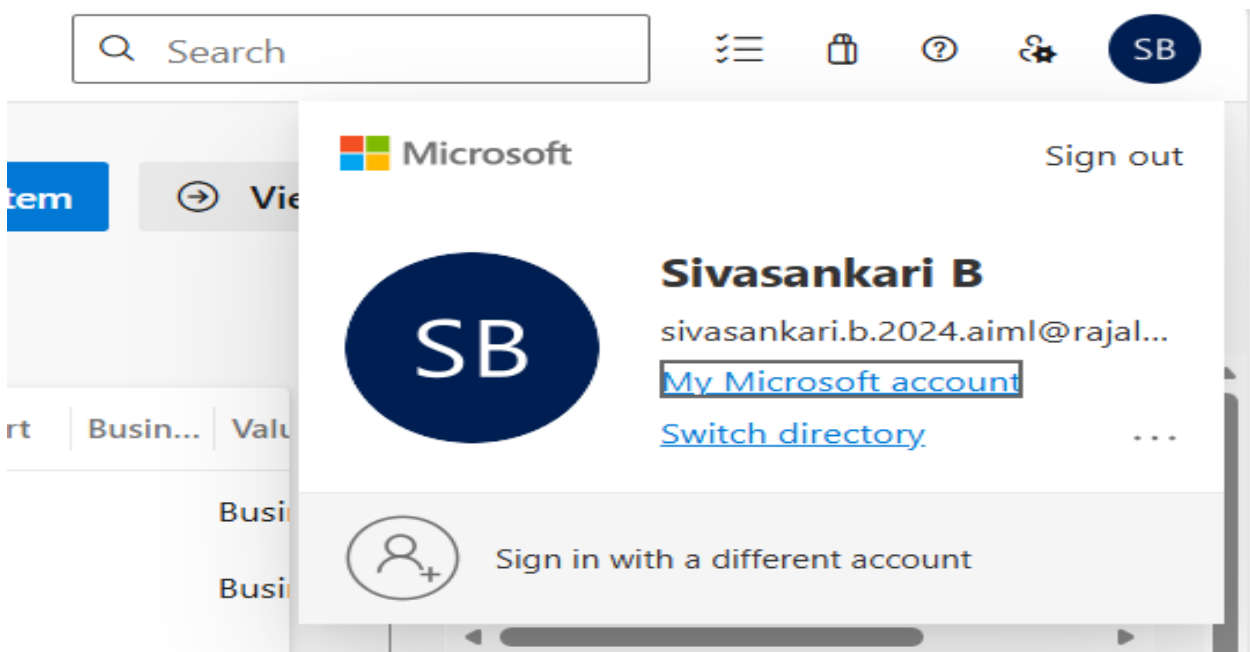
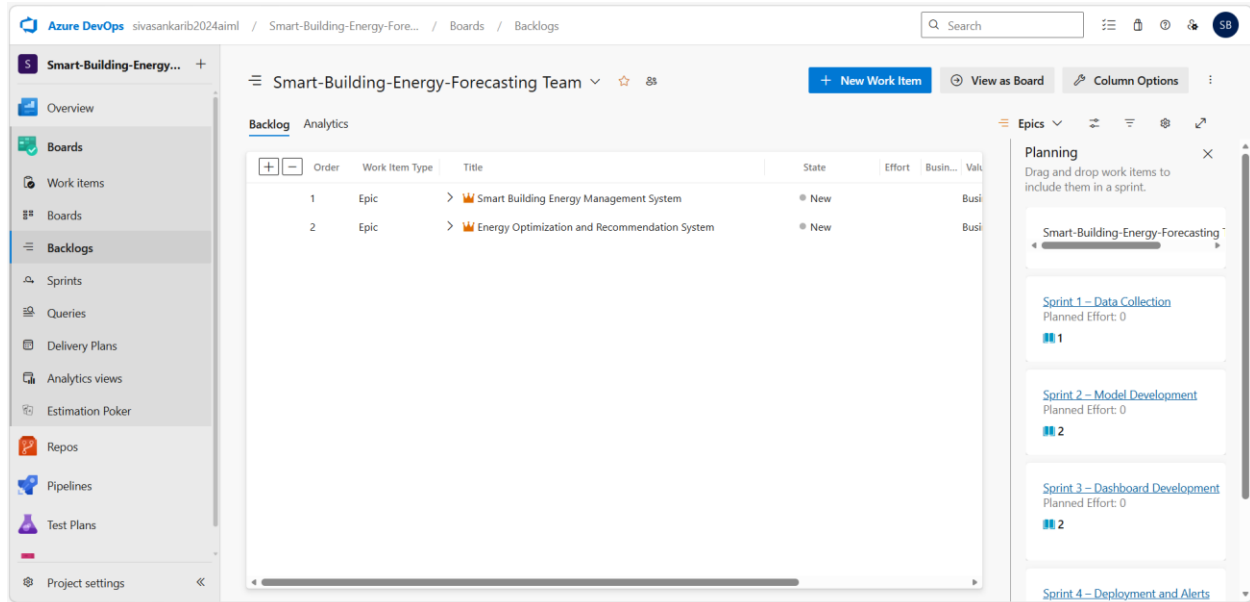


4. Project dashboard



5. To manage user stories:

- From the **left-hand navigation menu**, click on **Boards**. This will take you to the main **Boards** page, where you can manage work items, backlogs, and sprints.
- On the **work items** page, you'll see the option to **Add a work item** at the top. Alternatively, you can find a + button or **Add New Work Item** depending on the view you're in. From the **Add a work item** dropdown, select **User Story**. This will open a form to enter details for the new User Story.



Result:

Successfully created an Azure DevOps project with user story management and agile workflow setup.

EXP NO: 3

SETTING UP EPICS, FEATURES, AND USER STORIES FOR PROJECT PLANNING

Aim:

To learn about how to create epics, user story, features, backlogs for your assigned project.

Create Epic, Features, User Stories, Task

The screenshot shows the Azure DevOps interface for a team named 'Smart-Building-Energy-Forecasting Team'. The 'Backlog' view is active, displaying a list of work items. The left sidebar shows the navigation menu with 'Backlogs' selected. The main area shows a table of work items:

Order	Work Item Type	Title	State	Effort	Busin...	Valu...
1	Epic	Smart Building Energy Management System	New		Busi...	
	Feature	Smart Alerts	New		Busi...	
	User Story	As a system, I want to send alerts when energy consu...	New		Busi...	
2	Epic	Energy Optimization and Recommendation System	New		Busi...	
	Feature	Energy Data Collection	New		Busi...	
	User Story	Energy Data Collection	New		Busi...	
	User Story	As a user, I want to see energy usage charts so that I u...	New		Busi...	
	Feature	Energy Prediction System	New		Busi...	
	User Story	Energy Consumption Prediction	New		Busi...	

On the right, the 'Planning' sidebar is visible, showing a 'Smart-Building-Energy-Forecasting Team' backlog and three iterations (Iteration 1, Iteration 2, Iteration 3) with 'No work scheduled yet'.

1.Fill in Epics

The screenshot shows the Azure DevOps interface for a work item titled 'Smart Building Energy Management System'. The 'Work Items' view is active, displaying the details of the work item. The left sidebar shows the navigation menu with 'Work Items' selected. The main area shows the details of the work item:

Recently updated | [Back to Work Items](#) | 1 of 1

19 Smart Building Energy Management System

[gopika.v.2024.aiml@rajalaksmi.edu.](#) | 0 Comments | [Add Tag](#)

State: New | Area: Smart-Building-Energy-Forecasting | Reason: New | Iteration: Smart-Building-Energy-Forecasting

Description

System to monitor and predict energy consumption in smart buildings.

Discussion

Planning

Priority: 2
Risk:
Effort:
Business Value:
Time Criticality:
Start Date: Select a date...
Target Date: Select a date...

Deployment

To track releases associated with this work item, go to [Releases](#) and turn on deployment status reporting for Boards in your pipeline's Options menu. [Learn more about deployment status reporting](#)

Development

Add link

Link an Azure Repos [commit](#), [pull request](#) or [branch](#) to see the status of your development. You can also [create a branch](#) to get started.

Related Work

2.Fill in Features

The screenshot shows the 'Energy Prediction System' feature page in Azure DevOps. The breadcrumb trail is 'sivasankarib2024aiml / Smart-Building-Energy-Fore... / Boards / Work Items'. The feature is titled '22 Energy Prediction System' and is owned by 'Sivasankari B'. It has '0 Comments' and an 'Add Tag' button. The 'State' is 'New' and the 'Reason' is 'New'. The 'Area' and 'Iteration' are both 'Smart-Building-Energy-Forecasting'. The 'Description' field contains the text 'Predict future energy consumption using machine learning algorithms.' The 'Planning' section shows 'Priority' as '2', 'Risk' as 'Low', 'Effort' as 'Low', 'Business Value' as 'High', and 'Time Criticality' as 'High'. The 'Deployment' section has a 'To track releases associated with this work item, go to Releases and turn on deployment status reporting for Boards in your pipeline's Options menu. Learn more about deployment status reporting' message. The 'Development' section has an 'Add link' button and a message 'Link an Azure Repos commit, pull request or branch to see the status of your development. You can also create a branch to get started.' The 'Related Work' section is empty.

3.Fill in User Story Details

The screenshot shows the 'NEW USER STORY' page in Azure DevOps. The breadcrumb trail is 'sivasankarib2024aiml / Smart-Building-Energy-Fore... / Boards / Work Items'. The user story is titled 'As a user, I want to see energy usage charts so that I understand power consumption trends.' It has '0 Comments' and an 'Add Tag' button. The 'State' is 'New' and the 'Reason' is 'New'. The 'Area' and 'Iteration' are both 'Smart-Building-Energy-Forecasting'. The 'Description' field contains the text 'Click to add Description.' The 'Acceptance Criteria' field contains the text 'Click to add Acceptance Criteria.' The 'Discussion' field is empty. The 'Planning' section shows 'Story Points' as '1', 'Priority' as '2', and 'Risk' as 'Low'. The 'Classification' section shows 'Value area' as 'Business' and 'Business' as 'Business'. The 'Deployment' section has a 'To track releases associated with this work item, go to Releases and turn on deployment status reporting for Boards in your pipeline's Options menu. Learn more about deployment status reporting' message. The 'Development' section has an 'Add link' button and a message 'Link an Azure Repos commit, pull request or branch to see the status of your development. You can also create a branch to get started.' The 'Related Work' section is empty.

Result:

Thus, the creation of epics, features, user story and task has been created successfully.

EXP NO: 4

SPRINT PLANNING

Aim:

To assign user story to specific sprint for the Smart-Building-Energy-Forecasting project.

Sprint Planning Sprint 1

The screenshot shows the Azure DevOps Sprints board for the 'Smart-Building-Energy-Forecasting Team'. The left sidebar contains navigation options: Overview, Boards, Work items, Backlogs, Sprints (selected), Queries, Delivery Plans, Analytics views, Estimation Poker, Repos, Pipelines, Test Plans, and Project settings. The main area displays the 'Sprint 1 - Data Collection' board. The board has columns for 'New', 'Active', 'Resolved', and 'Closed'. A card titled '29 Data Collection' is in the 'New' column, with sub-items 'New' and 'Unassigned'. The board also shows 'Person: All' and 'No iteration dates Set dates'.

Sprint 2

The screenshot shows the Azure DevOps Sprints board for the 'Smart-Building-Energy-Forecasting Team'. The left sidebar is the same as in the previous screenshot. The main area displays the 'Sprint 2 - Model Development' board. The board has columns for 'New', 'Active', 'Resolved', and 'Closed'. Two cards are visible: '31 Energy Prediction Algorithm' (New) and '30 Machine Learning Model Development' (Active). The board also shows 'Person: All' and 'No iteration dates Set dates'.

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Sprint 3

The screenshot shows the Azure DevOps interface for the 'Smart-Building-Energy-Forecasting Team'. The left sidebar contains navigation options: Overview, Boards, Work Items, Backlogs, Sprints (selected), Queries, Delivery Plans, Analytics views, Estimation Poker, Repos, Pipelines, and Test Plans. The main area displays the 'Sprint 3 - Dashboard Development' board. It features a Kanban-style layout with columns: New, Active, Resolved, and Closed. Two work items are visible in the 'New' column: '33 Visualization Charts' (assigned to Sivasankari B) and '32 Dashboard Development' (assigned to sreethej.s.2024.aiml@...). The board includes a search bar, a 'New Work Item' button, and a 'Column Options' menu. The URL at the bottom is <https://dev.azure.com/sivasankarib2024aiml/Smart-Building-Energy-Forecasting>.

Sprint 4

The screenshot shows the Azure DevOps interface for the 'Smart-Building-Energy-Forecasting Team'. The left sidebar is identical to the previous screenshot, with 'Sprints' selected. The main area displays the 'Sprint 4 - Deployment and Alerts' board. It features a Kanban-style layout with columns: New, Active, Resolved, and Closed. Three work items are visible in the 'New' column: '36 Testing' (assigned to gopika.v.2024.aiml@r...), '35 Deployment' (assigned to sreethej.s.2024.aiml@...), and '34 Alert System' (assigned to keerthana.v.2024.aiml...). The board includes a search bar, a 'New Work Item' button, and a 'Column Options' menu. The URL at the bottom is <https://dev.azure.com/sivasankarib2024aiml/Smart-Building-Energy-Forecasting>.

Result:

The Sprints are created for the Smart-Building-Energy-Forecasting project.

EXP NO: 5

POKER ESTIMATION

Aim:

Create Poker Estimation for the user stories - Smart-Building-Energy-Forecasting project.

Poker Estimation

USER STORY 26

26 As a user, I want to see energy usage charts so that I understand power consumption trends.

keerthana.v.2024.aiml@rajalakshmi

0 Comments

Add Tag

Save and Close

Follow

Updated by Sivasankari B: Just now

State

New

Area

Smart-Building-Energy-Forecasting

Reason

New

Iteration

Smart-Building-Energy-Forecasting

Details

1

0

Description

Click to add Description.

Acceptance Criteria

Click to add Acceptance Criteria.

Discussion

10

Add a comment. Use # to link a work item, @ to mention a person, or ! to link a pull request.

Planning

Story Points

3

Priority

2

Risk

Classification

Value area

Business

Deployment

To track releases associated with this work item, go to [Releases](#) and turn on deployment status reporting for Boards in your pipeline's Options menu. [Learn more about deployment status reporting](#)

Development

Add link

Link an Azure Repos [commit](#), [pull request](#) or [branch](#) to see the status of your development. You can also [create a branch](#) to get started.

Related Work

Result:

The Estimation/Story Points is created for the project using Poker Estimation.

STUDENT REG NO

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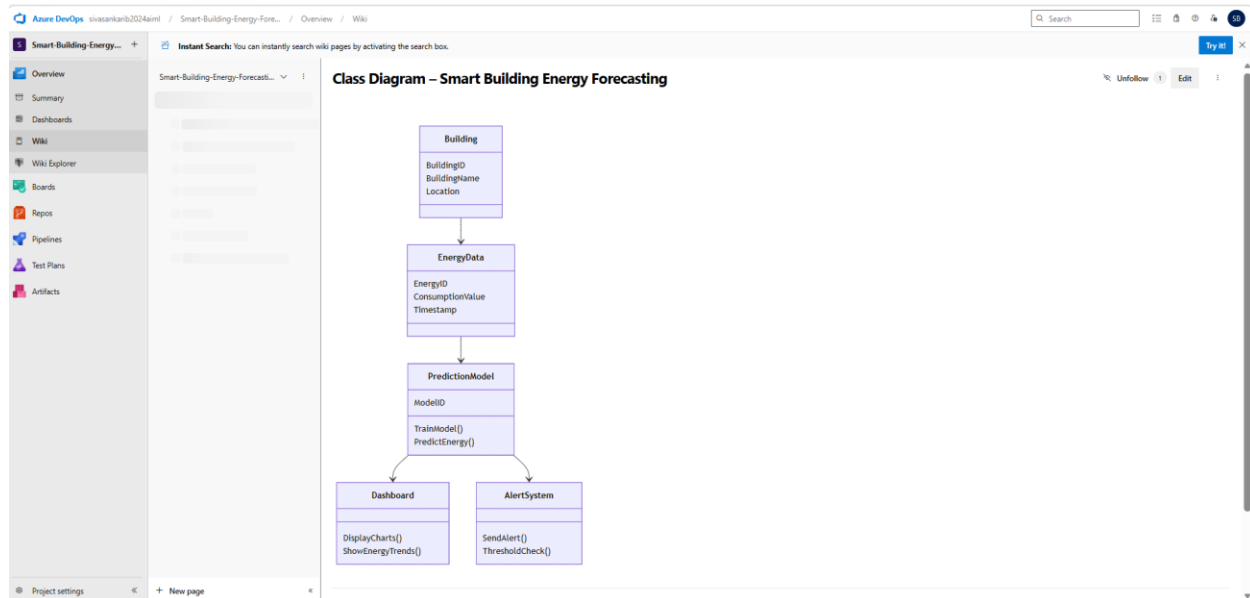
EXP NO: 6

DESIGNING CLASS AND SEQUENCE DIAGRAMS FOR PROJECT ARCHITECTURE

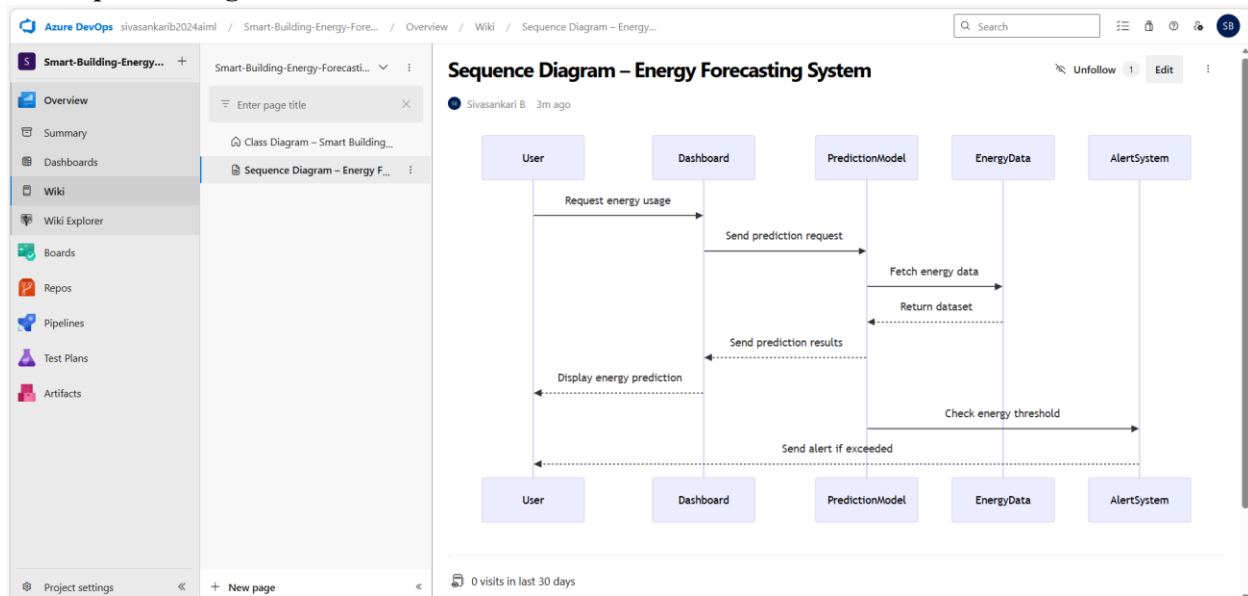
Aim:

To Design a Class Diagram and Sequence Diagram for the given Project.

6A. Class Diagram



6B. Sequence Diagram



Result:

The Class Diagram and Sequence Diagram is designed Successfully for the Smart-Building-Energy-Forecasting project.

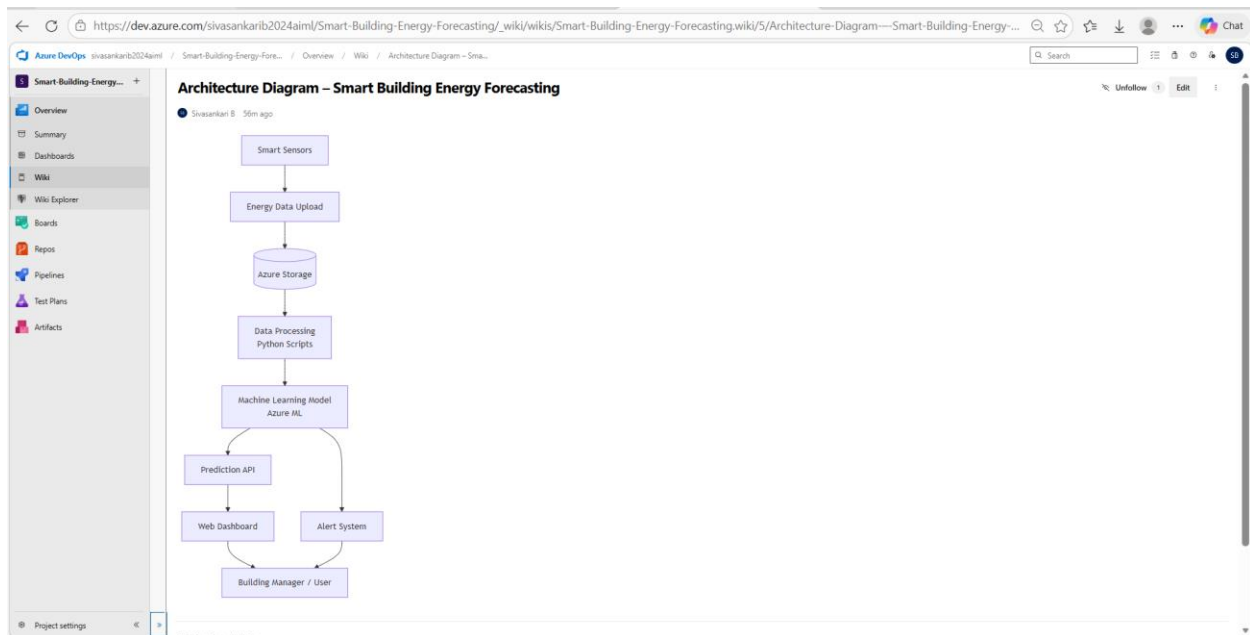
EXP NO: 7

DESIGNING ARCHITECTURAL AND ER DIAGRAMS FOR PROJECT STRUCTURE

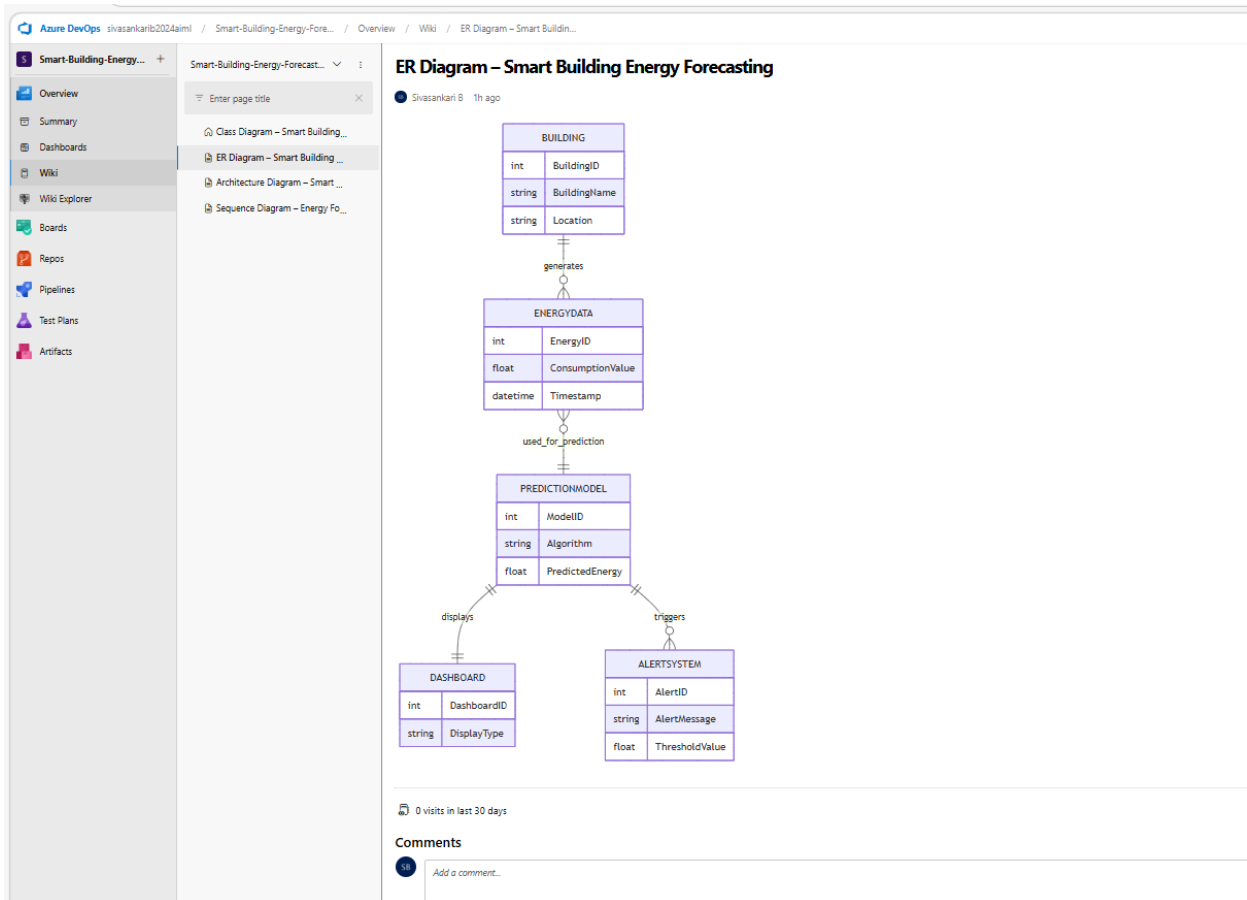
Aim:

To Design an Architectural Diagram and ER Diagram for the given Project.

7A. Architectural Diagram



7B.ER Diagram



Result:

The Architecture Diagram and ER Diagram is designed Successfully for the Smart-Building-Energy-Forecasting project.

EXP NO: 8	TESTING – TEST PLANS AND TEST CASES
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Aim:

Test Plans and Test Case and write two test cases for at least five user stories showcasing the happy path and error scenarios in azure DevOps platform.

Test Planning and Test Case

Test Case Design Procedure

1. Understand Core Features of the Application

- User Signup & Login
- Viewing and Managing Dashboard
- Fetching Real-time Metadata
- Editing playlists (rename, reorder, record)
- Creating smart audio playlists based on categories.

2. Define User Interactions

- Each test case simulates a real user behaviour (e.g., logging in, playlist, sending alert).

3. Design Happy Path Test Cases

- Focused on validating that all features function as expected under normal conditions.
- Example: User logs in successfully, adds item to playlist, or creates a category-based playlist.

4. Design Error Path Test Cases

- Simulate negative or unexpected scenarios to test robustness and error handling.
- Example: Login fails with invalid credentials, save fails when offline, no recommendations found.

5. Break Down Steps and Expected Results

- Each test case contains step-by-step actions and a corresponding expected outcome.
- Ensures clarity for both testers and automation scripts.

6. Use Clear Naming and IDs

- Test cases are named clearly .
- Helps in quick identification and linking to user stories or features.

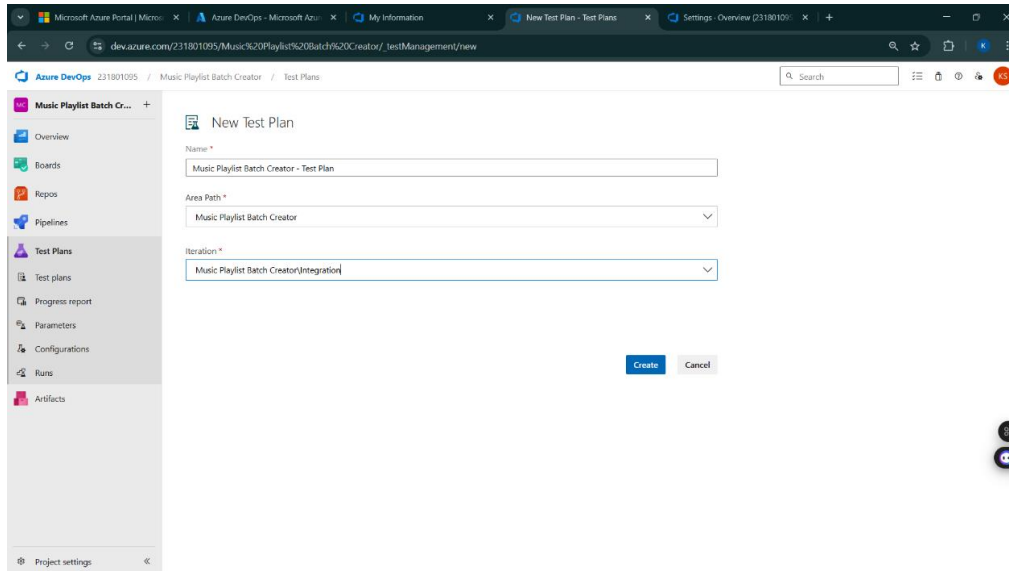
7. Separate Test Suites

- Grouped test cases based on functionality (e.g., Recommendation System).
- Improves organization and test execution flow in Azure DevOps.

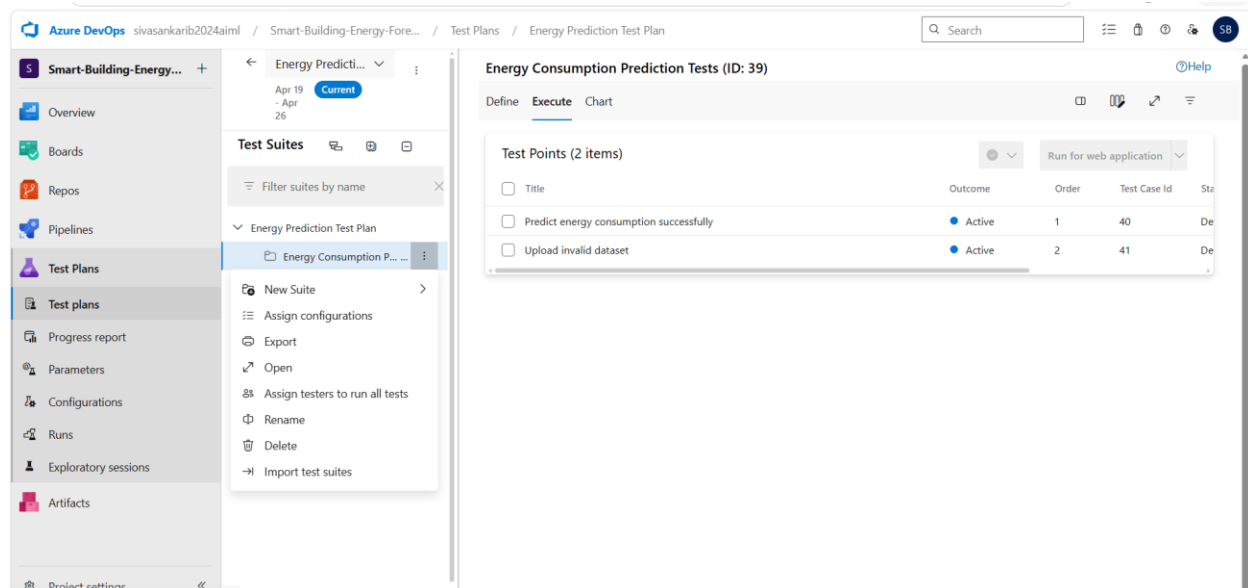
8. Prioritize and Review

- Critical user actions are marked high-priority.
- Reviewed for completeness and traceability against feature requirements.

1.New test plan



2.Test suite



3.Test case

Give two test cases for at least five user stories showcasing the happy path and error scenarios in azure DevOps platform.

Music Playlist Batch Creator – Test Plans

USER STORIES

- As a user, I want to sign up and log in securely so that I can access my playlists (ID: 79).
- As a user, I need to see my playlist in one place (ID: 76).
- As a user, I should be able to create an audio playlist as needed (ID: 73).
- As a user, I should be able to rename, record, and change the playlist (ID: 68).
- As a user, I need to have real-time metadata (ID: 65).

Test Suites

Test Suit: TS01 - User Login (ID: 86)

1. TC01 – Successful Sign Up

- **Action:**
 - Go to the Sign-Up page.
 - Enter valid name, email, and password.
 - Click "Sign Up".
- **Expected Results:**
 - Sign-Up form is displayed.
 - Fields accept values without error.
 - Account is created, and the user is redirected to the dashboard.
- **Type:** Happy Path

2. TC02 – Secure Login

- **Action:**
 - Go to the Login page.
 - Enter valid email and password.
 - Click on "Login".
- **Expected Results:**
 - Login form is displayed.
 - Fields accept data without error.
 - User is logged in and redirected to the dashboard.
- **Type:** Happy Path

3. TC03 – Sign Up with Existing Email

- **Action:**
 - Go to the Sign-Up page.
 - Enter a name and an already registered email.
 - Click on "Sign Up".
- **Expected Results:**
 - Fields accept data.

- Error message "Email already registered" is displayed.
- **Type:** Error Path

4. TC04 – Login with Wrong Password

- **Action:**
 - Go to the Login page.
 - Enter valid email and incorrect password.
 - Click on "Login".
- **Expected Results:**
 - Input is accepted.
 - Error message "Invalid username or password" is shown.
- **Type:** Error Path

Test Suit: TS02 - View Playlists (ID: 87)

1. TC05 – View Playlist Page

- **Action:**
 - Log in successfully.
 - Navigate to "My Playlists" section.
- **Expected Results:**
 - All created playlists are displayed clearly.
- **Type:** Happy Path

2. TC06 – Playlist Loading Failure

- **Action:**
 - Disconnect from the internet.
 - Navigate to "My Playlists".
- **Expected Results:**
 - Network is offline.
 - Error message "Unable to load playlists" is shown.
- **Type:** Error Path

Test Suit: TS03 - Real-Time Metadata (ID: 88)

1. TC07 – Real-Time Metadata Display

- **Action:**
 - Play a song.
 - Observe the metadata panel.
- **Expected Results:**
 - Metadata (title, artist, album, duration) is displayed and updates in real time.
- **Type:** Happy Path

2. TC08 – Metadata Not Updating

- **Action:**
 - Play a different song.

- Observe the metadata panel.
- **Expected Results:**
 - Metadata remains static or shows default/fallback message.
- **Type:** Error Path

Test Suit: TS04 - Playlist Editing (ID: 89)

1. TC09 – Rename Playlist Successfully

- **Action:**
 - Navigate to "My Playlists".
 - Click "Rename" next to a playlist.
 - Enter a new name and click "Save".
- **Expected Results:**
 - Playlist name updates successfully.
- **Type:** Happy Path

2. TC10 – Rename with Blank Name

- **Action:**
 - Click "Rename" on a playlist.
 - Leave the field blank.
 - Click "Save".
- **Expected Results:**
 - Error message "Playlist name cannot be empty" is shown.
- **Type:** Error Path

3. TC11 – Change Playlist Order

- **Action:**
 - Open a playlist.
 - Drag and drop songs to reorder.
 - Click "Save".
- **Expected Results:**
 - Playlist order is updated and saved.
- **Type:** Happy Path

4. TC12 – Change Playlist Order Fails

- **Action:**
 - Login and go to "My Playlists".
 - Select a playlist.
 - Go offline or simulate server error.
 - Reorder songs and click "Save Order".
- **Expected Results:**
 - Error message: "Failed to update order. Please check your connection".
- **Type:** Error Path

Test Suit: TS05 - Smart Playlist Creation (ID: 90)

1. TC13 – Generate Playlist Based on Various Categories

- **Action:**
 - Login with valid credentials.
 - Click on "Generate Playlist".
 - Select categories.
 - Click "Generate Playlist".
- **Expected Results:**
 - Playlist is generated based on selected mood and categories.
- **Type:** Happy Path

2. TC14 – Fail to Generate Playlist Due to Missing Category Selection or Invalid Input

- **Action:**
 - Login with valid credentials.
 - Click on "Generate Playlist".
 - Select categories.
 - Click "Generate Playlist".
- **Expected Results:**
 - Error message: "Please select at least one valid category" or "No recommendations found for the selected filters".
- **Type:** Error Path

Test Cases

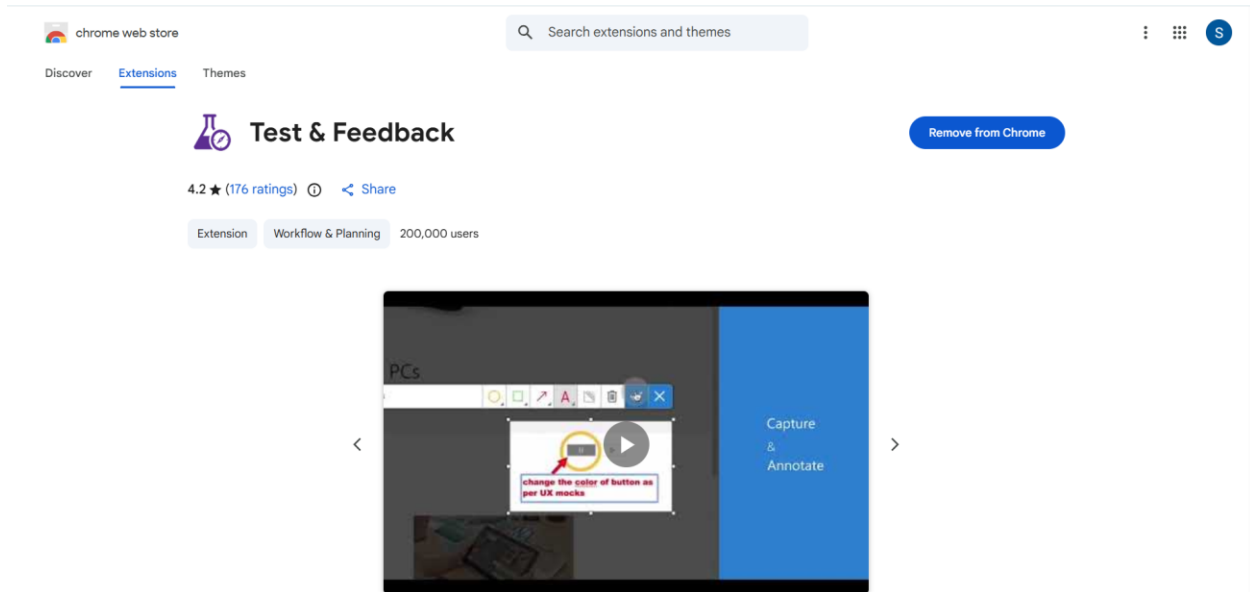
The screenshot displays a test case management interface. At the top, the test case is titled "TEST CASE 40: Predict energy consumption successfully" and is associated with user "Sivasankari B". It has 0 comments and tags. The test case is in the "Design" state, with a reason of "New". It belongs to the "Smart-Building-Energy-Forecasting" area and the "Smart-Building-Energy-Forecasting\Sprint 4 – Deployment and Alerts" iteration. The test case was last updated by Sivasankari B 50 minutes ago.

The "Steps" section contains a table with two steps:

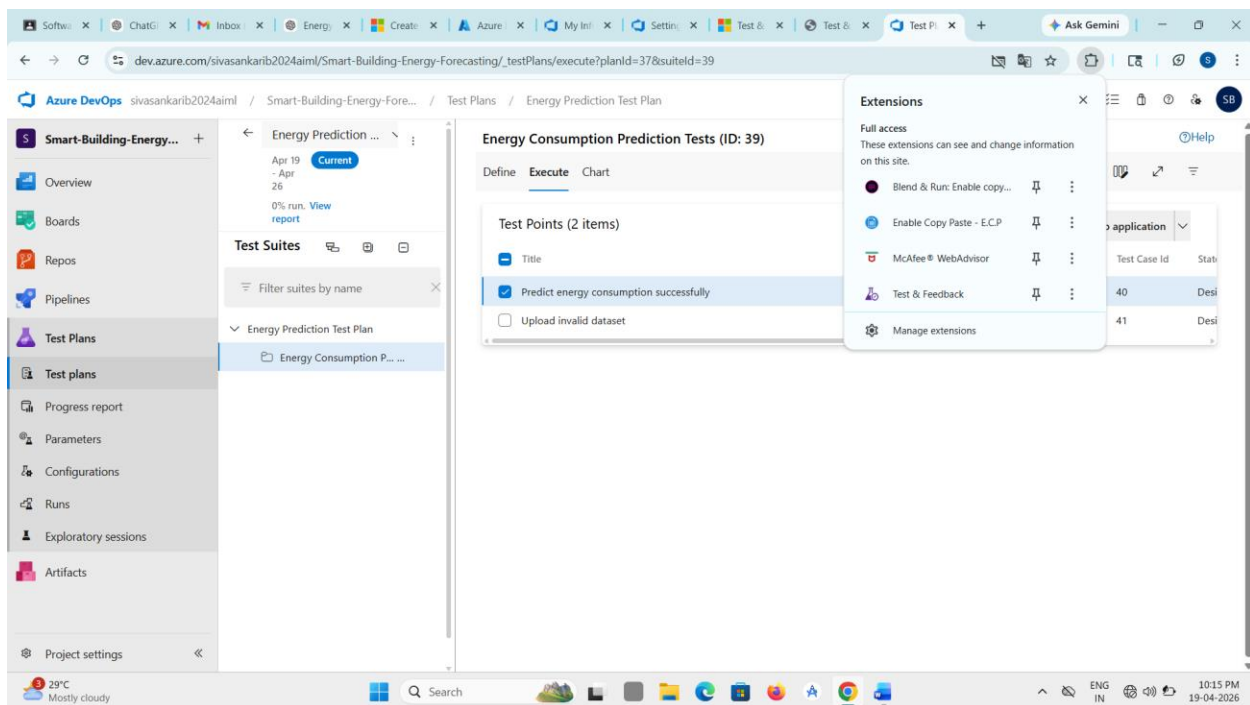
Steps	Action	Expected result	Atte
1.	Upload energy dataset	Dataset uploaded successfully	
2.	Run prediction model	System generates predicted energy usage	

Below the table, there is a prompt: "Click or type here to add a step".

The right sidebar shows "Recent test results" with a link to "Run manual tests with web runner. Learn how to get started". Below this is the "Deployment" section, which provides instructions on tracking releases and links to "Releases" and "deployment status reporting". The "Development" section includes an "Add link" button and a prompt to "Link an Azure Repos commit, pull request or branch to".



Test and feedback
Showing it as an extension



5. Running the test cases

dev.azure.com/sivasankarib2024aiml/Smart-Building-Energy-Forecasting/_testPlans/execute?planId=37&suiteId=39

Azure DevOps sivasankarib2024aiml / Smart-Building-Energy-Fore... / Test Plans / Energy Prediction Test Plan

Energy Consumption Prediction Tests (ID: 39)

Define Execute Chart

Test Points (2 items)

Title	Outcome	Order	Test Case Id	Status
☒ Predict energy consumption successfully	In Progress	1	40	Desi
☐ Upload invalid dataset	Active	2	41	Desi

View execution history
Mark Outcome
Run
Reset test to active
Edit test case
Assign tester
View test result

Run for web application
Run for desktop application
Run with options

Runner - Test Plans - Google Chrome

dev.azure.com/sivasankarib2024aiml/Smart-Building-Energy-Forecasting/_testExecution/Index

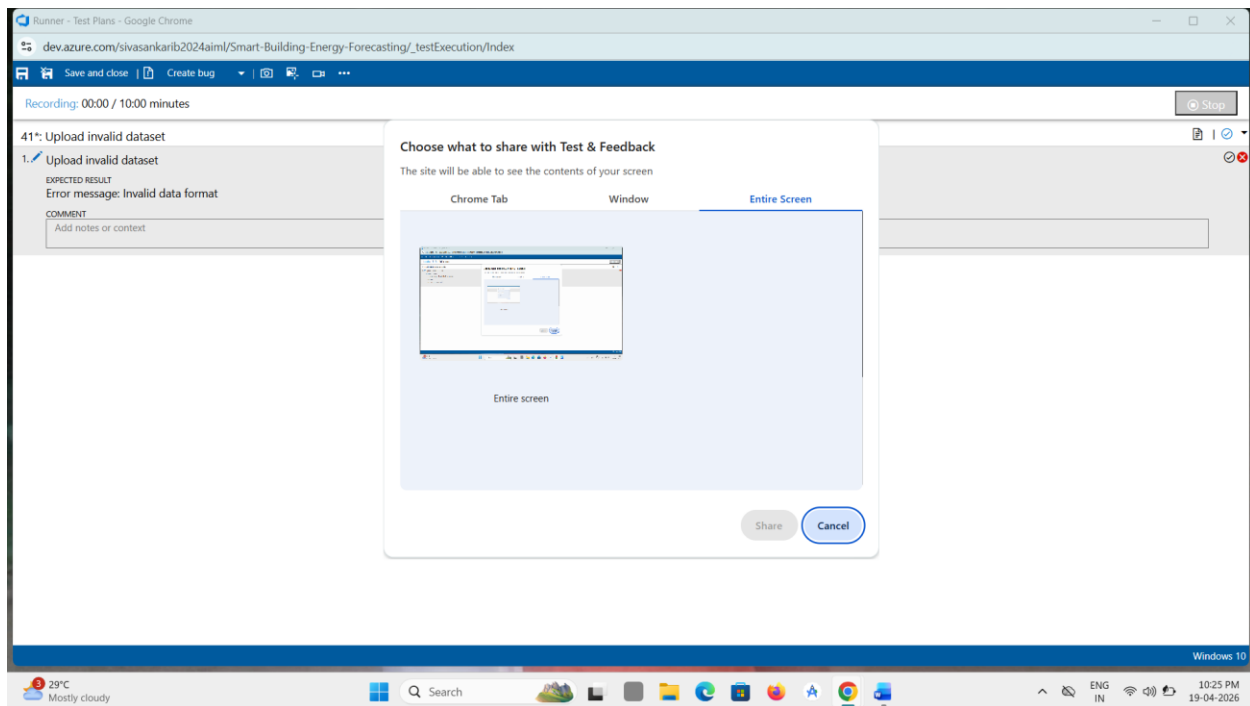
40*: Predict energy consumption successfully

1. Upload energy dataset
EXPECTED RESULT
Dataset uploaded successfully

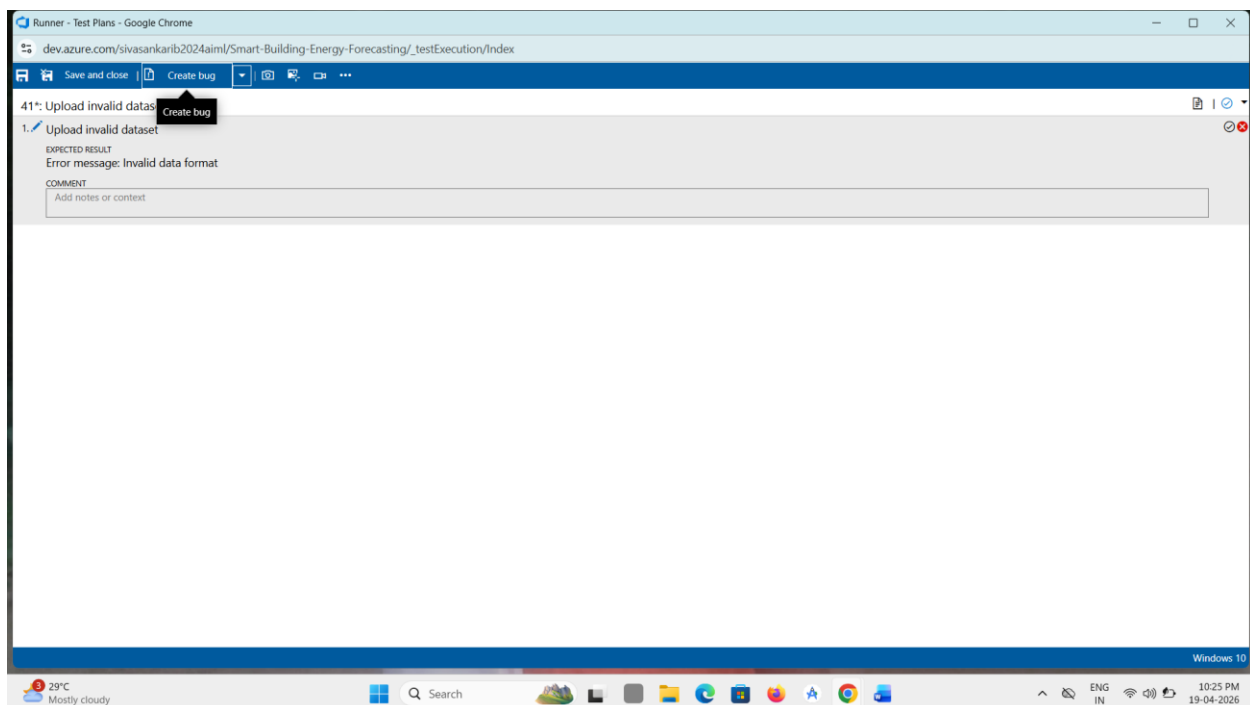
2. Run prediction model
EXPECTED RESULT
System generates predicted energy usage

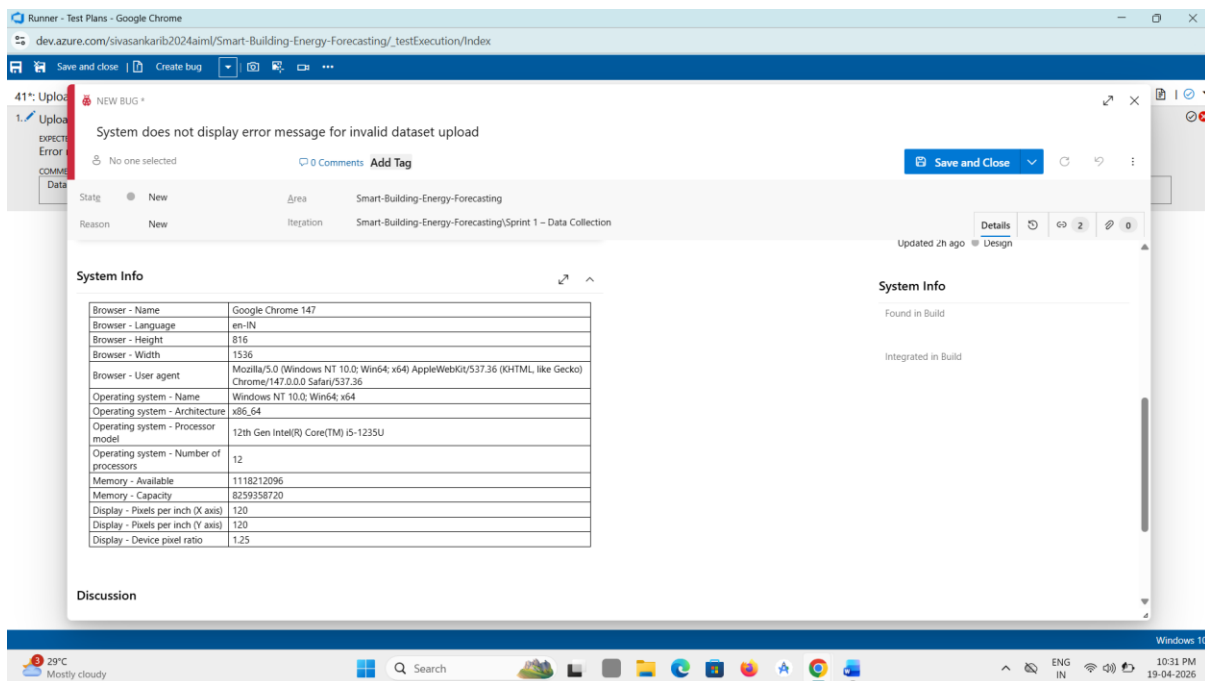
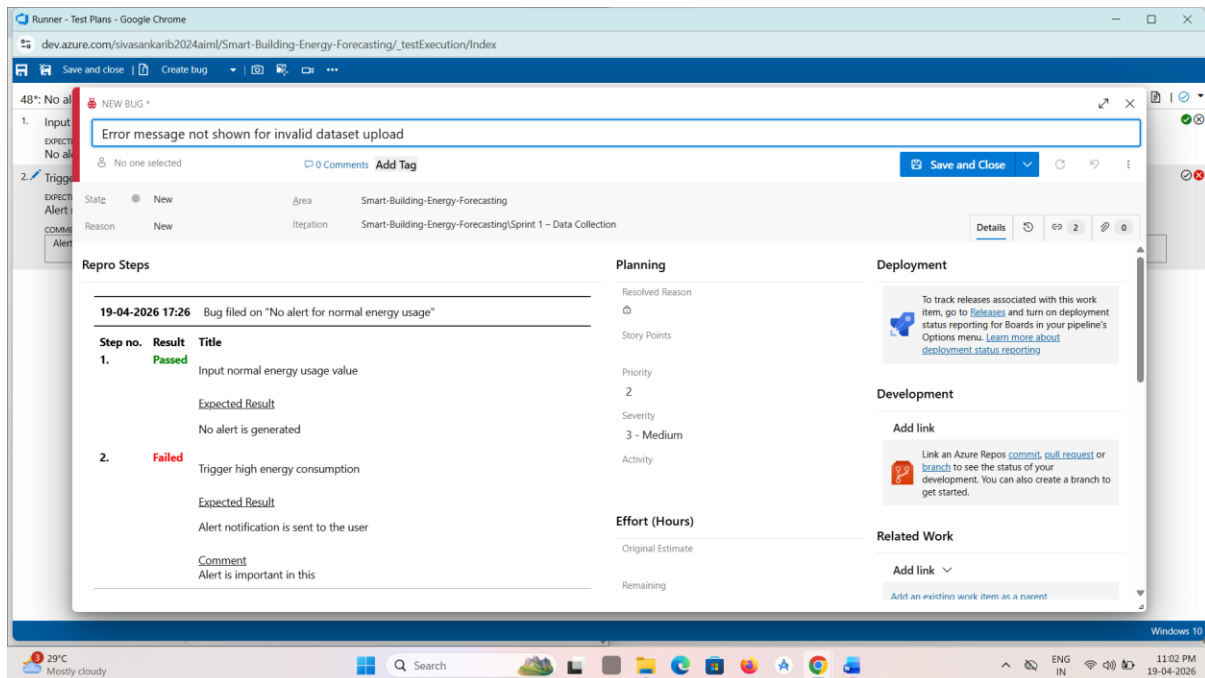
Pass test step (Press T + P)

6. Recording the test case



7. Creating the bug





8. Test case results

The screenshot shows the Azure DevOps Test Plans interface for a project named 'Smart-Building-Energy-Fore...'. The left sidebar contains navigation options: Overview, Boards, Repos, Pipelines, Test Plans (selected), Progress report, Parameters, Configurations, Runs, Exploratory sessions, and Artifacts. The main area displays the 'Energy Prediction Test Plan' with a 'Current' status and a progress bar showing '50% run, 100% passed'. Below this, a list of 'Test Suites' is shown, including 'TS 3-Alert System Tes...', 'TS 2-Dashboard Testi...', and 'TS 1-Energy Consumpt...'. The right pane shows the 'Test Points (2 items)' for 'TS 3-Alert System Testing (ID: 46)'. The table below lists the test points:

Title	Outcome	Order	Test Case Id	Status
☒ Generate alert when energy exceeds threshold	Passed	1	47	Desi
☐ No alert for normal energy usage	Failed	2	48	Desi

9. Test report summary

The screenshot shows the Azure DevOps Work Items interface for a project named 'Smart-Building-Energy-Fore...'. The left sidebar contains navigation options: Overview, Boards, Work items (selected), Backlogs, Sprints, Queries, Delivery Plans, Analytics views, Estimation Poker, Repos, Pipelines, Test Plans, and Project settings. The main area displays a 'Bug 50' with the title 'No message sent for users'. The 'State' is 'New' and the 'Reason' is 'New'. The 'Area' is 'Smart-Building-Energy-Forecasting' and the 'Iteration' is 'Smart-Building-Energy-Forecasting/Sprint 1 - Data Collection'. The 'Repro Step' section shows the following steps:

Step no.	Result	Title
1.	Passed	Input normal energy usage value
		Expected Result
		No alert is generated
2.	Failed	Trigger high energy consumption
		Expected Result
		Alert notification is sent to the user
		Comment
		Alert is important in this

The 'Planning' section shows the 'Resolved Reason' as 'Story Points', 'Priority' as '2', 'Severity' as '3 - Medium', and 'Activity' as 'Effort (Hours)'. The 'Deployment' section shows a link to 'Releases' and a link to 'Deployment status reporting'. The 'Development' section shows a link to 'Add link' and a link to 'Add link'.

- Assigning bug to the developer and changing state

Bug 50 No message sent for users

gopika.v.2024.ai@rajalaksmi.edu 0 Comments Add Tag

State: New Area: Smart-Building-Energy-Forecasting Reason: New Iteration: Smart-Building-Energy-Forecasting/Sprint 1 – Data Collection

Updated by Sivasankari B: 23m ago

Repro Steps

19-04-2026 17:35 Bug filed on "No alert for normal energy usage"

Step no.	Result	Title
1.	Passed	Input normal energy usage value <u>Expected Result</u> No alert is generated
2.	Failed	Trigger high energy consumption <u>Expected Result</u> Alert notification is sent to the user <u>Comment</u> Alert is important in this

Planning

Resolved Reason

Story Points

Priority: 2

Severity: 3 - Medium

Activity

Effort (Hours)

Original Estimate

Deployment

To track releases associated with this work item, go to [Releases](#) and turn on deployment status reporting for Boards in your pipeline's Options menu. [Learn more about deployment status reporting](#)

Development

Add link

Link an Azure Repos [commit](#), [pull request](#) or [branch](#) to see the status of your development. You can also [create a branch](#) to get started.

Related Work

10. Progress report

Summary

1 Test plans 6 Test points

6 (6 / 6) Test points run

100% Run

100% (6 / 6) Pass rate

6 Passed

Outcome trend

Last 14 Days

Test

0 2 4 6

2026-04-05 2026-04-06 2026-04-07 2026-04-08 2026-04-09 2026-04-10 2026-04-11 2026-04-12 2026-04-13 2026-04-14 2026-04-15 2026-04-16 2026-04-17 2026-04-18 2026-04-19

Passed

Details

Test plan name	Test points	Run %	Passed %	Failed %	Not run count
Energy Prediction Test Plan	6	100	100	0	0
TS 1-Energy Consumption Prediction Tests	2	100	100	0	0
TS 2-Dashboard Testing	2	100	100	0	0
TS 3-Alert System Testing	2	100	100	0	0

11.Changing the test template

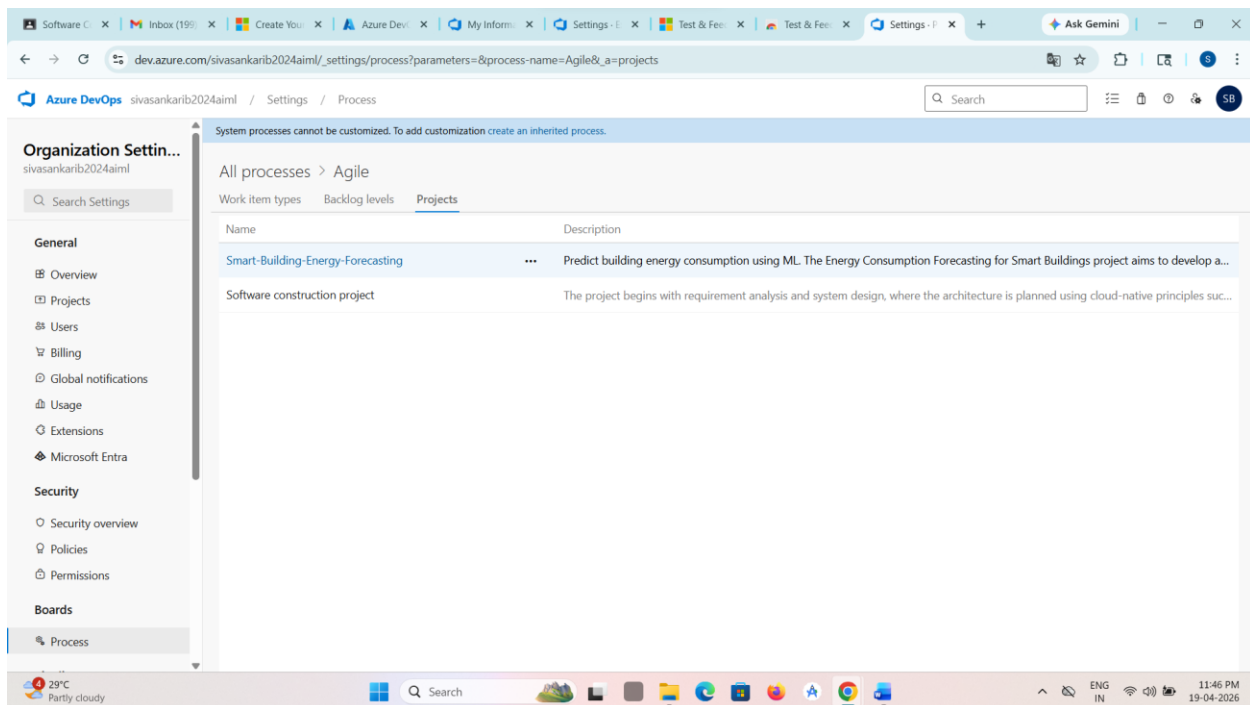
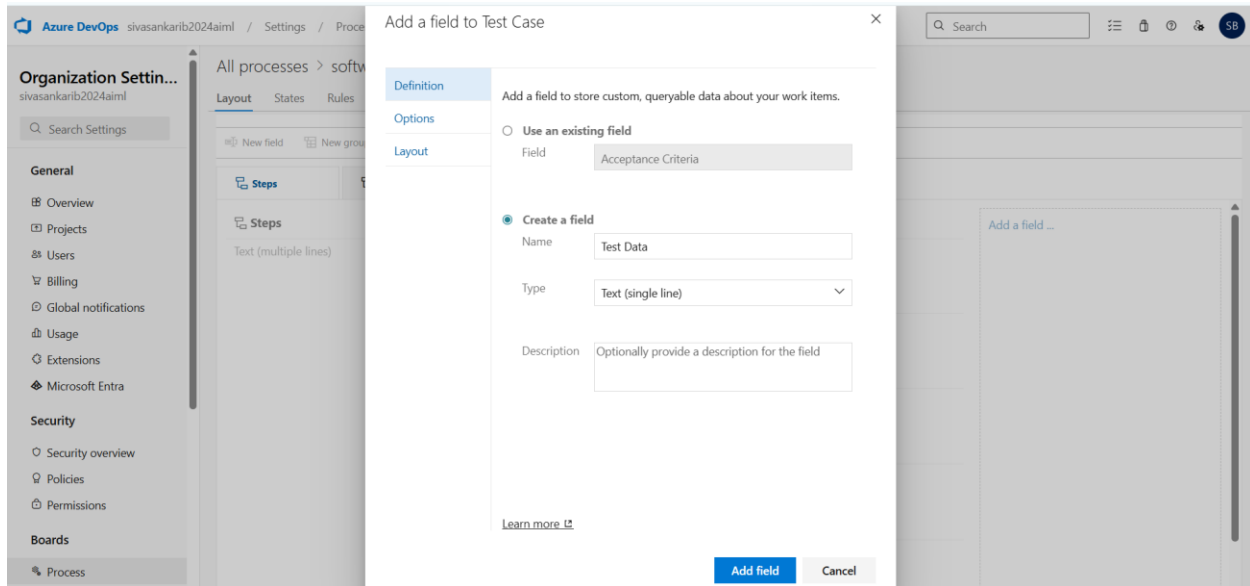
The screenshot shows the Azure DevOps 'All processes' page. The left sidebar contains the 'Organization Settings' menu with options like General, Overview, Projects, Users, Billing, Global notifications, Usage, Extensions, Microsoft Entra, Security, and Boards. The 'Process' option is selected. The main area displays a table of processes:

Name	Description	Team projects
Basic (default)	This template is flexible for any process and great for teams getting started with Az...	0
Agile	This template is flexible and will work great for most teams using Agile planning me...	2
Scrum	This template is for teams who follow the Scrum framework.	0
CMMI	This template is for more formal projects requiring a framework for process improv...	0

The screenshot shows the Azure DevOps 'All processes' page. The left sidebar is the same as the previous image. The main area displays a table of processes:

Name	Description	Team projects
Basic (default)	This template is flexible for any process and great for teams getting started with Az...	0
Agile	This template is flexible and will work great for most teams using Agile planning me...	2
software construction project		0
Scrum	This template is for teams who follow the Scrum framework.	0
CMMI	This template is for more formal projects requiring a framework for process improv...	0

12.View the new test case template



Organization Settin...

sivasankarib2024aiml

Search Settings

General

Overview

Projects

Users

Billing

Global notifications

Usage

Extensions

Microsoft Entra

Security

Security overview

Policies

Permissions

Boards

Process

All processes > software construction project > Test Case

Layout States Rules

New field

New group

New page

Get extensions

Steps

Summary

Associated Aut...

Steps

Text (multiple lines)

Recent test results

Recent test case results

Deployment

Deployments

Development

Links

Related Work

Links

Status

Priority

Integer

Custom

Test Data

Text (single line)

Result:

The test plans and test cases for the user stories is created in Azure DevOps with Happy Path and Error Path

EXP NO: 9

LOAD TESTING AND PIPELINES

Aim:

To create an Azure Load Testing resource and run a load test to evaluate the performance of a target endpoint and to create and demonstrate an Azure DevOps pipeline for automating application builds, tests, and deployment.

Load Testing

Azure Load Testing:

Azure Load Testing allows you to simulate high traffic and stress tests for your web applications and APIs to understand how they perform under load. It helps identify performance bottlenecks, scalability issues, and optimize resource usage before deployment.

Steps to Create an Azure Load Testing Resource:

Before you run your first test, you need to create the Azure Load Testing resource:

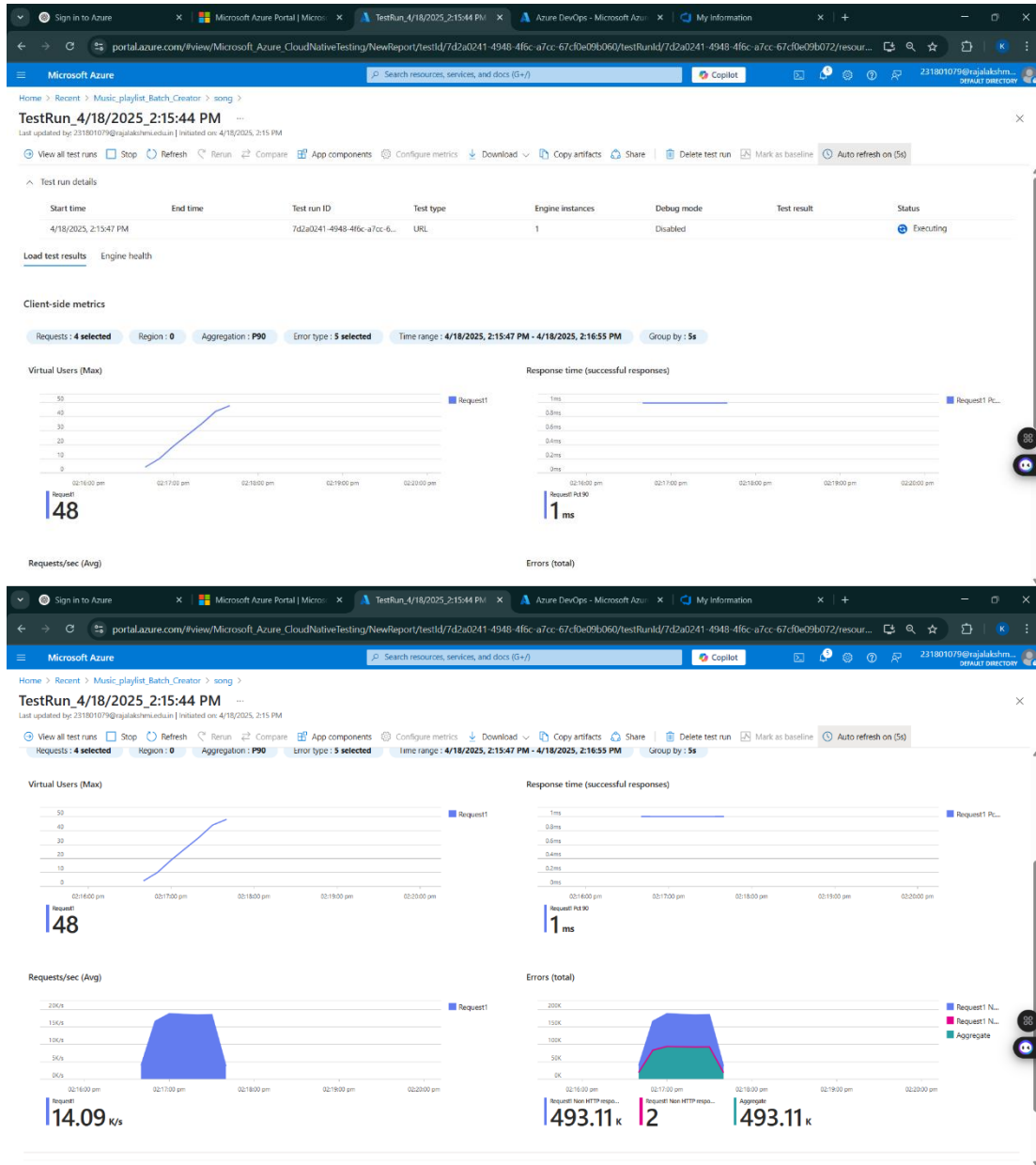
1. Sign in to Azure Portal
Go to <https://portal.azure.com> and log in.
2. Create the Resource
 - Go to *Create a resource* → Search for “Azure Load Testing”.
 - Select Azure Load Testing and click Create.
3. Fill in the Configuration Details
 - *Subscription*: Choose your Azure subscription.
 - *Resource Group*: Create new or select an existing one.
 - *Name*: Provide a unique name (no special characters).
 - *Location*: Choose the region for hosting the resource.
4. (Optional) Configure tags for categorization and billing.
5. Click Review + Create, then Create.
6. Once deployment is complete, click Go to resource.

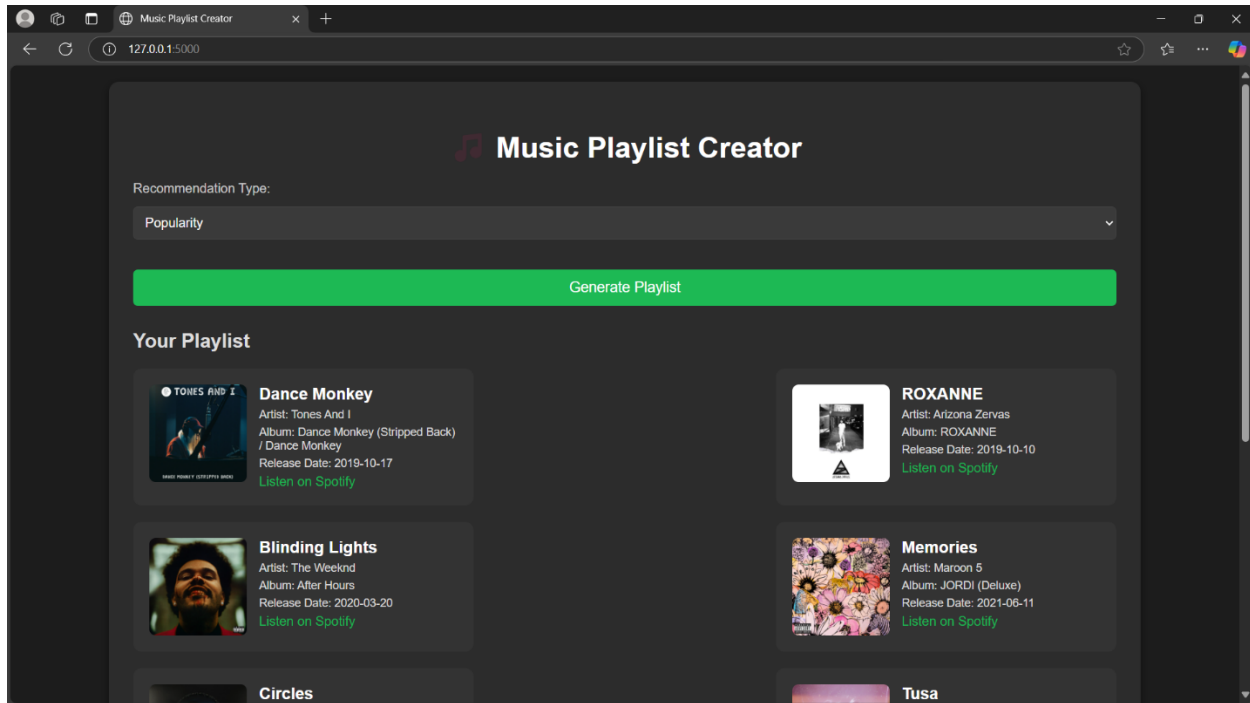
Steps to Create and Run a Load Test:

Once your resource is ready:

1. Go to your Azure Load Testing resource and click Add HTTP requests > Create.
2. Basics Tab
 - *Test Name*: Provide a unique name.
 - *Description*: (Optional) Add test purpose.
 - *Run After Creation*: Keep checked.
3. Load Settings
 - *Test URL*: Enter the target endpoint (e.g., <https://yourapi.com/products>).
4. Click Review + Create → Create to start the test.

Load Testing





Pipelines

Description:

This experiment demonstrates how to connect a GitHub-hosted Flask-based music recommendation project with Azure DevOps. The pipeline will automatically install dependencies, run basic tests, and publish artifacts. This ensures that every commit triggers checks for reliability and smooth deployment.

Steps:

1. Connect GitHub to Azure DevOps:
 - In Azure DevOps, create a new project.
 - Create a pipeline and select GitHub as the source.
 - Authorize access to your GitHub repository, ensuring that Azure DevOps can pull the repository for your pipeline.
2. Create azure-pipelines.yml in Your Repo Root:
 - In your GitHub repository, create a new file called azure-pipelines.yml in the root directory.
 - Add the following basic pipeline configuration for Python and Flask:

yml Code

trigger:

- main # Trigger pipeline when changes are pushed to the main branch

pool:

vmImage: ubuntu-latest # Use a hosted Ubuntu agent

steps:

Step 1: Checkout the code from GitHub

- checkout: self

Step 2: Set up Python environment

- task: UsePythonVersion@0

inputs:

versionSpec: '3.x' # Use the latest Python 3.x version

displayName: "Set up Python"

Step 3: Install dependencies from the correct path

- script: |

python -m pip install --upgrade pip

pip install -r project/requirements.txt # Adjusted path to requirements.txt

displayName: "Install dependencies"

Step 4: Run a simple Python script to check the environment

- script: |

python -c "print('🎵 Hello from Music Playlist Batch Creator!')"

displayName: "Run a Python script"

3. Pipeline Tasks Include:

- Setting up the Python environment using the UsePythonVersion task.
- Installing project dependencies from project/requirements.txt. Make sure the path to requirements.txt is correct (it is located under the project folder).
- Running a simple Python script to verify that Python is set up correctly and the pipeline works.

4. Run and Monitor Pipeline:

- Commit changes to the main branch of your repository to trigger the pipeline in Azure DevOps.
- Monitor the logs in the Azure DevOps portal to view logs, errors, or success messages and ensure everything runs smoothly.

Pipeline

The screenshot displays the Azure DevOps web interface for a pipeline run. The browser address bar shows the URL: `dev.azure.com/231801095/Music%20Playlist%20Batch%20Creator/_build/results?buildId=308&view=results`. The left sidebar contains navigation links: Overview, Boards, Repos, Pipelines (selected), Environments, Releases, Library, Task groups, Deployment groups, Test Plans, and Artifacts. The main content area shows the details for Pipeline 2, run #20250424.3. A message states: "This run is being retained as one of 3 recent runs by main (Branch)." with a "View retention leases" link. Below this, the "Summary" tab is active, showing it was manually run by Karthick S. The repository is "Music Playlist Batch Creator" on the "main" branch with commit "a87bd670". The run started "Just now" and took "24s". It has "0 work items" and "0 artifacts". A "View 52 changes" link is present. The "Jobs" section shows a single job named "Job" with a status of "Success" and a duration of "6s". The bottom of the sidebar shows "Project settings".

#20250424.3 • Pipeline 2
Music Playlist Batch Creator (9)

This run is being retained as one of 3 recent runs by main (Branch). [View retention leases](#)

Summary Code Coverage

Manually run by **Karthick S** [View 52 changes](#)

Repository and version: Music Playlist Batch Creator
main a87bd670

Time started and elapsed: Just now 24s

Related: 0 work items 0 artifacts

Tests and coverage: [Get started](#)

Jobs

Name	Status	Duration
Job	Success	6s

Result:

Successfully created the Azure Load Testing resource and executed a load test to assess the performance of the specified endpoint and also demonstrated pipelines in azure devops.

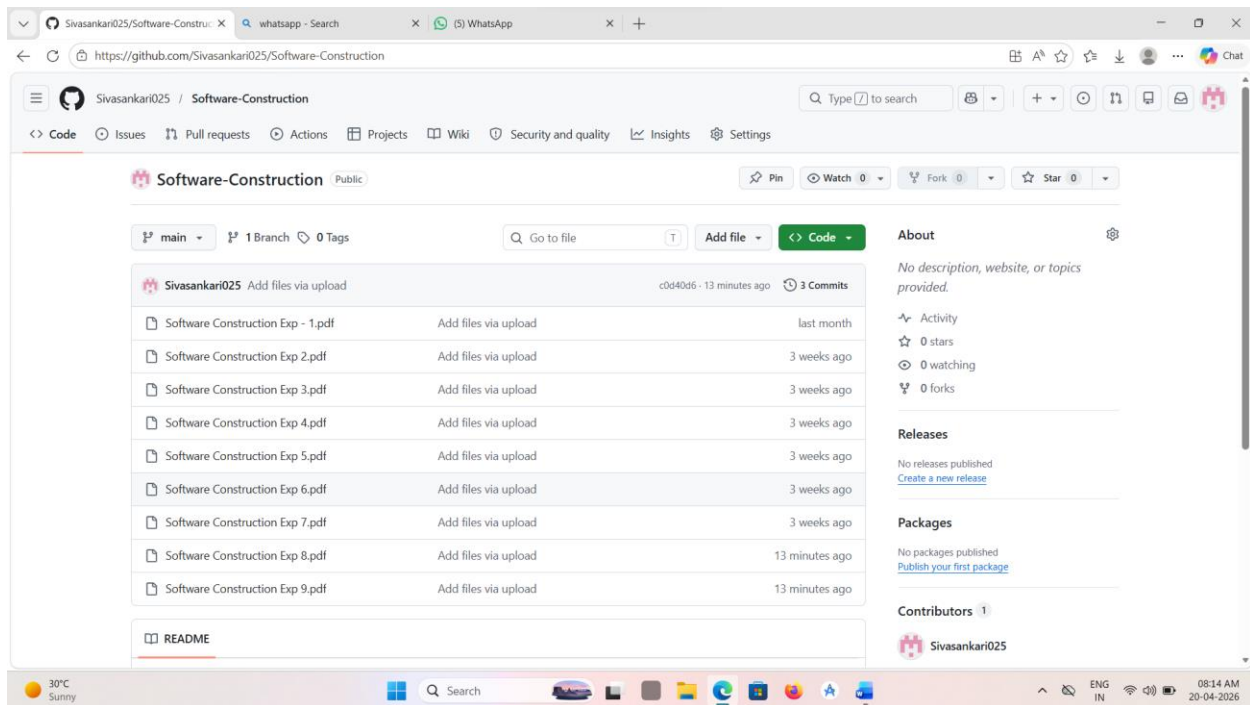
EXP NO: 10

GITHUB: PROJECT STRUCTURE & NAMING CONVENTIONS

Aim:

To provide a clear and organized view of the project's folder structure and file naming conventions, helping contributors and users easily understand, navigate, and extend the Smart-Building-Energy-Forecasting project.

GitHub Project Structure



Result:

The GitHub repository clearly displays the organized project structure and consistent naming conventions, making it easy for users and contributors to understand and navigate the codebase.